

# SHUVAM PAL

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## Education

**Dayananda Sagar College of Engineering**

Bengaluru, India

*B.E. Artificial Intelligence and Machine Learning*

## Experience

**Bentham AI** 

Dec 2025 – Feb 2026

*Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions*

*Remote*

- Built a resilient **Puppeteer/Node.js automation engine** for MCA filings, reducing manual filing time by **85%** through retries, network-idle monitoring, and recoverable browser-state handling.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized services with **Docker**, deployed to **Google Cloud Run**, and automated releases with **GitHub Actions**, improving deployment velocity by **70%**.

**Google Summer of Code** 

May 2025 – Sept 2025

*Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX*

*Remote*

- Built MRI segmentation tooling for **95 anatomical regions** and integrated model outputs into visualization workflows that needed repeatable, inspectable processing.
- Profiled volume-processing paths and reduced runtime by **87%** with async/parallel execution and fewer repeated post-processing passes.
- Improved 3D reconstructed surface visualization flows, keeping image-processing outputs usable for debugging and review.

## Open Source

[Contributions](#) 

- Contributed to Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, adapting fixes to maintainer review and existing observability/debugging practices.

## Projects

**Postificus** | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, AWS, Prometheus*

[GitHub](#) 

- Built observable Go services for a **content distribution platform** using browser automation, with RabbitMQ workers, DLQs, retries, Redis autosave, and PostgreSQL persistence.
- Added circuit-breaker logic and failure isolation for third-party publishing paths, keeping browser automation failures observable and recoverable instead of cascading through the system.

**Swish Support System** | *Python, Go, FastAPI, Gin, PostgreSQL, Redis, Langfuse*

[GitHub](#) 

- Built an AI support system with per-conversation context for active item, issue type, requested resolution, pending confirmations, and evidence artifacts, achieving **85%+** policy-correct resolutions on a **50-case** eval set.
- Reduced compensation leakage by **25%** versus a naive refund baseline by separating semantic understanding from deterministic policy and tracing high-risk cases with **Langfuse**.

**Murdoc** | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

[GitHub](#) 

- Instrumented an AI gateway with **OpenTelemetry metrics/traces**, Prometheus health checks, route profiles, and audit logs tying model/tool requests to policy outcomes.
- Built replayable decision records for prompt, tool, retrieval, and output layers so high-risk agent behavior could be correlated with request context and failure modes.

## Technical Skills

**Languages:** Go, Python, TypeScript

**Frameworks:** FastAPI, Echo, Gin, Node.js

**Tools:** OpenTelemetry, Prometheus, Langfuse, PostgreSQL, Redis, RabbitMQ, Docker, GitHub Actions